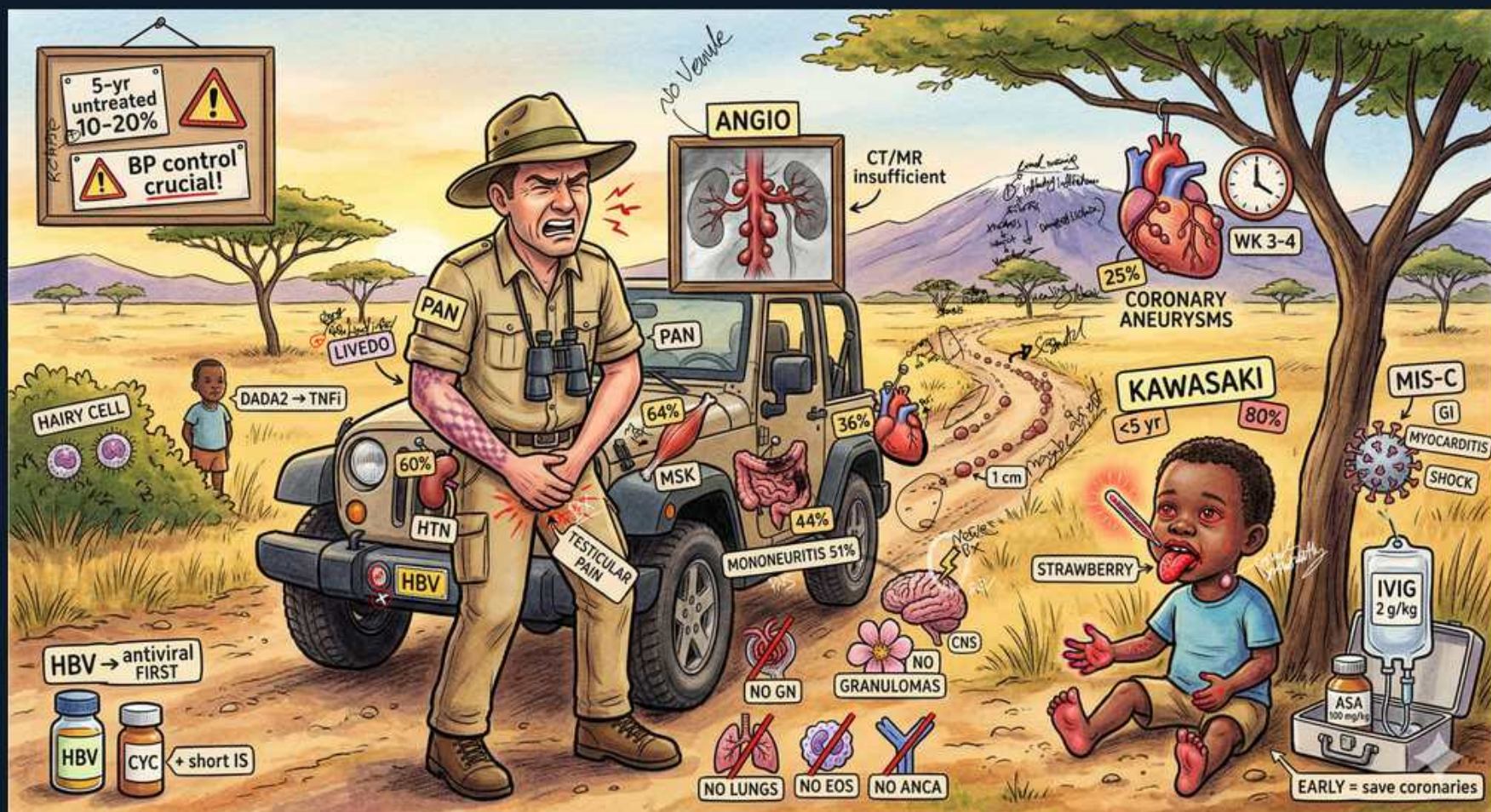


# Vasculitis — PAN (Medium Vessel) & Kawasaki





1. PAN safari figure

WHAT YOU SEE

Khaki-clad man labeled PAN beside a jeep on the savanna

WHAT IT ENCODES

Polyarteritis nodosa is a necrotizing vasculitis of MEDIUM-sized muscular arteries, sparing the smallest vessels (arterioles, capillaries, venules). It presents with constitutional symptoms plus multi-organ ischemia: renal (microaneurysms, infarcts, renovascular HTN), GI (mesenteric angina, bowel infarction), skin (livedo, nodules, ulcers), nerves (mononeuritis multiplex), and testicles. Mnemonic: PAN spares the Pulmonary And glomerular (no lungs, no GN).

BOARD TRAP

PAN is medium-vessel and ANCA-NEGATIVE — a vignette with palpable purpura + glomerulonephritis + alveolar hemorrhage is small-vessel/ANCA disease (MPA/GPA), not PAN. Choosing PAN for a pulmonary-renal syndrome is the trap; PAN spares both lungs and glomeruli.

2. Renal angiography — microaneurysms

WHAT YOU SEE

Kidney angiogram with ANGIO label, CT/MR insufficient note

WHAT IT ENCODES

PAN renal disease is VASCULAR, not glomerular: catheter angiography of renal and mesenteric arteries shows microaneurysms and alternating stenoses, plus cortical infarcts. Conventional angiography is more sensitive than CT/MR angiography for these small microaneurysms. Diagnosis rests on angiography and/or biopsy of an affected accessible site (skin, nerve/muscle, testis).

BOARD TRAP

Renal involvement in PAN is ischemic (microaneurysms/infarcts, renovascular HTN), NOT glomerulonephritis. Expecting RBC casts/crescentic GN points to ANCA vasculitis instead. Also, choosing renal BIOPSY for suspected PAN risks bleeding from microaneurysms — angiography is preferred.

3. Bead-like aneurysms

WHAT YOU SEE

Beaded vessel trail across the road, 1 cm beads

WHAT IT ENCODES

The hallmark angiographic finding is 'beads on a string' — alternating saccular microaneurysms (often ~1 cm) and segmental stenoses along medium muscular arteries, classically in renal and mesenteric beds. These aneurysms can rupture, causing retroperitoneal or intra-abdominal hemorrhage. Histology shows segmental transmural necrotizing inflammation with fibrinoid necrosis at vessel bifurcations.

BOARD TRAP

'Beading' microaneurysms are characteristic of PAN but not unique — they also occur in segmental arterial mediolysis, fibromuscular dysplasia, and amphetamine/drug-induced vasculopathy. The discriminator is the systemic vasculitic picture (mononeuritis, livedo, HTN, HBV link) plus biopsy.

4. HBV — antiviral first

WHAT YOU SEE

HBV antiviral FIRST, HBV + CYC + short IS tags bottom left

WHAT IT ENCODES

A significant minority of PAN is hepatitis B-associated (immune-complex driven by HBV antigenemia). Treat HBV-PAN with antiviral therapy (e.g., entecavir/tenofovir) PLUS a SHORT course of glucocorticoids and plasma exchange to clear immune complexes — minimize/avoid prolonged immunosuppression, which promotes viral replication. Idiopathic (non-HBV) PAN is treated with steroids ± cyclophosphamide for severe disease.

BOARD TRAP

Giving prolonged steroids/cyclophosphamide ALONE for HBV-associated PAN is the trap — unopposed immunosuppression drives HBV replication and worsens outcomes. Antiviral therapy must come first (with plasma exchange + a short steroid course).

5. HTN (renin)

WHAT YOU SEE

An HTN 60% tag on the safari figure — renin-driven hypertension from renal-artery vasculitis hits roughly 60% of PAN patients.

WHAT IT ENCODES

PAN causes renovascular (renin-mediated) hypertension from renal artery involvement and parenchymal infarcts in up to ~60% of patients. Blood-pressure control is a treatment priority because uncontrolled HTN drives renal and cardiovascular morbidity. ACE inhibitors/ARBs are useful given the renin-angiotensin contribution.

BOARD TRAP

New severe hypertension in a patient with myalgias, neuropathy, and skin nodules is a clue to PAN, not 'essential HTN.' Treating the BP alone without recognizing the underlying vasculitis is the trap.

6. Mononeuritis multiplex 51%

WHAT YOU SEE

MONONEURITIS 51% tag

WHAT IT ENCODES

Mononeuritis multiplex — asymmetric, painful, stepwise involvement of named peripheral nerves (foot drop from common peroneal, wrist drop, etc.) — occurs in roughly half of PAN cases from ischemia of the vasa nervorum. Nerve/muscle (e.g., sural nerve + gastrocnemius) biopsy is a high-yield, accessible way to confirm the diagnosis. It is one of the most characteristic PAN features.

BOARD TRAP

Mononeuritis multiplex is non-specific (also EGPA, MPA, GPA, RA, cryoglobulinemia, diabetes). The discriminator for PAN is the accompanying medium-vessel picture (renovascular HTN, GI angina, testicular pain, HBV) with NO lung/GN involvement and a negative ANCA.



7. Livedo / skin

WHAT YOU SEE

A LIVEDO tag on the figure's mottled, net-patterned skin — the lacy livedo reticularis of medium-vessel skin involvement.

WHAT IT ENCODES

Cutaneous PAN manifestations: livedo reticularis/racemosa, tender subcutaneous nodules along arteries, digital ischemia/gangrene, and punched-out ulcers. A skin biopsy must be deep (reaching the dermal-subcutaneous junction) to capture the medium-sized arteries; a superficial punch biopsy may miss the diagnosis. Cutaneous-limited PAN is a milder subset confined to skin.

BOARD TRAP

A shallow/superficial skin biopsy can miss medium-vessel PAN because the affected arteries sit at the deep dermis/subcutis. Calling a biopsy 'negative for vasculitis' when it was too superficial is the trap — request a deep incisional/excisional biopsy.

8. Testicular / MSK

WHAT YOU SEE

TESTICULAR PAIN and MSK 64% tags on the jeep — testicular-artery vasculitis (a classic PAN clue) plus the very common myalgias/arthralgias.

WHAT IT ENCODES

Testicular pain/orchitis from testicular artery vasculitis is a classic, relatively specific PAN clue. Musculoskeletal symptoms — myalgias and arthralgias — are very common (~64%). Testicular involvement, when present and accessible, can even be a biopsy target.

BOARD TRAP

Unexplained testicular pain plus systemic symptoms (fever, weight loss, neuropathy, HTN) should prompt PAN, not just a urologic/infectious workup. Attributing it solely to epididymo-orchitis and missing systemic vasculitis is the trap.

9. No GN / No lungs

WHAT YOU SEE

NO GN, NO LUNGS, NO GRANULOMAS tags bottom

WHAT IT ENCODES

PAN characteristically SPARES the lungs (no pulmonary capillaritis/hemorrhage) and does NOT cause glomerulonephritis or form granulomas — because it affects medium arteries, not the small vessels/capillaries where GN and alveolar hemorrhage arise. These three 'NOs' separate PAN from ANCA-associated small-vessel vasculitis.

BOARD TRAP

Pulmonary involvement or glomerulonephritis essentially EXCLUDES PAN — those point to ANCA-associated vasculitis (MPA/GPA). Diagnosing PAN in a patient with hemoptysis/alveolar hemorrhage or crescentic GN is the trap.

10. No EOS / No ANCA

WHAT YOU SEE

NO EOS, NO ANCA tags

WHAT IT ENCODES

PAN is ANCA-NEGATIVE and is not an eosinophilic disease — it lacks the eosinophilia/asthma of EGPA and the ANCA of GPA/MPA. Diagnosis is made by angiography (beading microaneurysms) and/or deep tissue biopsy, supported by HBV serology. A positive ANCA should redirect you toward GPA/MPA/EGPA.

BOARD TRAP

A positive ANCA argues AGAINST PAN — choosing PAN in an ANCA-positive vignette is the trap. Likewise, marked eosinophilia + asthma points to EGPA, not PAN.

11. DADA2 — TNFi

WHAT YOU SEE

A DADA2 -> TNFi tag near a child figure — the monogenic PAN-mimic in the young whose strokes are prevented by TNF inhibitors, not cyclophosphamide.

WHAT IT ENCODES

DADA2 (Deficiency of Adenosine Deaminase 2) is an autosomal-recessive monogenic disease (ADA2/CECR1 mutation) presenting as PAN-like medium-vessel vasculitis in children/young adults, with early strokes (lacunar), livedo racemosa, fevers, and immunodeficiency/cytopenias. TNF inhibitors prevent strokes and are the treatment of choice. Suspect it in young/familial 'PAN' with recurrent strokes.

BOARD TRAP

Treating young-onset, stroke-prone, familial PAN-like vasculitis with standard cyclophosphamide while missing DADA2 is the trap — TNF inhibitors are the effective therapy, and anticoagulation/antiplatelets can cause hemorrhage in DADA2. Genetic testing changes management.

12. 5-yr untreated 10-20%

WHAT YOU SEE

5-yr untreated 10-20%, BP control crucial banner

WHAT IT ENCODES

Untreated PAN has a dismal prognosis (historically ~10-20% 5-year survival), dramatically improved by glucocorticoids ± cyclophosphamide (and antivirals for HBV-PAN). Blood-pressure control is crucial given renovascular hypertension. The Five-Factor Score (renal insufficiency, proteinuria, GI, cardiac, CNS involvement) stratifies severity and guides whether to add cyclophosphamide.

BOARD TRAP

PAN is potentially fatal if untreated, so 'observe and monitor' for organ-threatening disease is the trap. Severe PAN (high Five-Factor Score) requires prompt steroids PLUS cyclophosphamide, not steroids alone.

13. Kawasaki child / strawberry

WHAT YOU SEE

Child labeled KAWASAKI <5 yr holding strawberry, 80%

WHAT IT ENCODES

Kawasaki disease: medium-vessel vasculitis in children <5 yr (peak ~80% under 5). Diagnosis: fever ≥5 days PLUS ≥4 of — bilateral non-exudative conjunctivitis, oral changes (strawberry tongue, cracked red lips), polymorphous rash, extremity changes (palm/sole erythema/edema then periungual desquamation), and unilateral cervical lymphadenopathy (CRASH and Burn mnemonic).

BOARD TRAP

Incomplete/atypical Kawasaki (fever + only 2-3 criteria) still occurs, especially in infants who paradoxically have the HIGHEST coronary aneurysm risk. Withholding treatment because full criteria aren't met — when inflammatory markers and echo support it — is the trap.



## 14. Coronary aneurysms / IVIG

### WHAT YOU SEE

Heart with 25% CORONARY ANEURYSMS, WK 3-4, IVIG 2 g/kg + ASA

### WHAT IT ENCODES

Untreated Kawasaki leads to coronary artery aneurysms in ~20-25%, peaking at weeks 3-4, risking thrombosis, MI, and sudden death. Treatment: IVIG 2 g/kg single infusion + aspirin, given EARLY (ideally within 10 days of fever onset) to cut aneurysm risk to ~3-5%. Follow with serial echocardiography; IVIG-resistant cases may need a second IVIG dose, steroids, or infliximab.

### BOARD TRAP

Delaying IVIG beyond day 10 markedly reduces its protective effect on coronaries — the danger is waiting. Also note Kawasaki is the exception where aspirin IS used in a child (anti-inflammatory then antiplatelet doses), despite usual Reye-syndrome caution.

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## 15. MIS-C

### WHAT YOU SEE

MIS-C tag with GI, myocarditis, shock right side

### WHAT IT ENCODES

Multisystem Inflammatory Syndrome in Children (MIS-C) is a post-SARS-CoV-2 hyperinflammatory, Kawasaki-like illness, typically 2-6 weeks after infection, in older children than classic Kawasaki. It features prominent GI symptoms, myocarditis/ventricular dysfunction, and shock, with very high inflammatory markers and frequent positive COVID serology. Treatment overlaps Kawasaki: IVIG ± glucocorticoids, with cardiac support.

### BOARD TRAP

MIS-C skews to OLDER children with prominent GI symptoms, myocardial dysfunction, and shock — calling it 'classic Kawasaki' or missing the cardiac/shock component is the trap. Check COVID serology/exposure and obtain echo + troponin/BNP early.

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